

Operator Discovery Report: transfer_scaffold_mix_bottleneck_cluster_grid_safe

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic scaffold choice using instance descriptors: prefer nearest-neighbor 2-opt for clustered/high transfer stability; prefer cheapest-insertion 2-opt for two-cluster bottlenecks; prefer random-restart 2-opt for
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:nearest_neighbor_2opt:random_restart_2opt.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, clustered_tsp, grid_like_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: 2e318d0281e149da.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.3987.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 25.028343.

Transplant Results

- nearest_neighbor_2opt gap delta -0.068763, runtime delta 3994.265471, distance-eval delta 6276598.714286.
- cheapest_insertion_2opt gap delta 0.038559, runtime delta 547.690871, distance-eval delta -484459.285714.
- random_restart_2opt gap delta -0.048437, runtime delta 3806.556314, distance-eval delta 6262764.571429.
- sparse_three_opt gap delta -0.067477, runtime delta 3971.568986, distance-eval delta 6273177.428571.
- clustered_local_search gap delta -0.090286, runtime delta 3781.584557, distance-eval delta 6261929.285714.

Instance-Family Results

- heldout_tsplib operator gap 0.166295 vs baseline 0.235058.
- clustered_tsp operator gap 0.0 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.006647 vs baseline 0.003204.
- grid_like_tsp operator gap 0.095808 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.3987 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.001685 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.003539.
- Runtime inflation: 25.028343.
- Distance-evaluation inflation: 638.855145.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.003539.
- flattened_selector mean gap 0.121317 and delta vs full operator 0.003539.
- shuffled_selector mean gap 0.116775 and delta vs full operator -0.001003.